The Intertemporal Approach to the Current Account

1MT495 Current Issues in Applied International Economics

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Where We Are

Lectures 1–2

The accounting: $CA = S - I$ is net lending to the rest of the world, and it changes the investment position

Lectures 3–4

Static trade theory: *what* countries trade – by comparative advantage and by scale. Trade is balanced in every period

Today

Trade *through time*: why countries run current account surpluses and deficits at all

After today you should be able to:

1. state the intertemporal budget constraint and explain what it implies for a debtor country
2. solve the two-period small open economy model and show the current account in a diagram
3. predict the current account response to temporary, permanent and anticipated shocks, and to investment opportunities
4. explain how the world interest rate is set when large countries trade over time
5. use the model to read real cases – oil shocks, Norway, Israel, the US and China – and explain where it fails

Outline

1. From Accounting to Theory
2. The Two-Period Small Open Economy
3. Investment, the Long Run and the World Interest Rate
4. Applications and Evidence
5. Critiques: A Real Model in a Financial World
6. Conclusions

Reminder: What Does the Balance of Payments Measure?

Transactions – flows

The balance of payments records transactions between residents and non-residents during a period. The **current account** records trade and income; the **financial account** records net acquisition of foreign assets and incurrence of liabilities.

Positions – stocks

The international investment position records the existing claims on, and obligations to, the rest of the world. Its change also reflects **valuation and other changes**, not just new transactions.

$$CA + KA + NEO = FA \quad \text{(BPM6 net-acquisition sign convention)}$$

Takeaway: A current account deficit describes a net resource transfer. To understand financial risks, also examine the gross transactions and balance-sheet positions behind it.

Current Account vs. Financial Financing: One Example

Suppose an economy buys 100 of foreign goods today and exports only 80. Ignore income transfers, capital-account entries and valuation changes.

Real side

$$TB = 80 - 100 = -20$$

The country absorbs 20 more goods than it produces for the rest of the world. Its net foreign asset position falls by 20.

Financial side

$$\Delta B = CA = -20$$

Foreign investors could buy domestic claims, or residents could sell foreign assets. The underlying gross flows can be much larger than 20.

Takeaway: Borrowing is the way current consumption can exceed current resources. The CA records the net outcome, not the entire financing chain.

Recap: The Identities From Lectures 1 and 2

Balance of payments (BPM6):

$$CA + KA + NEO = FA$$

Current account:

$$CA = TB + \text{primary income} + \text{secondary income}$$

Saving and investment:

$$CA = S - I = (S_P - I_P) + (S_G - I_G)$$

The stock:

$$\Delta NIIP = CA + KA + NEO + \text{valuation} + \text{other}$$

In today's models (no capital account, no valuation): net foreign assets B change only through the current account:

$$CA_t = B_t - B_{t-1} = TB_t + rB_{t-1}$$

Takeaway: The identities say what must add up. Today's question is behavioural: *why* do households, firms and governments choose to save more or less than they invest?

What Is the Intertemporal Approach?

- Views the current account as the result of **forward-looking saving and investment decisions** (Sachs 1981; Obstfeld and Rogoff 1995, 1996)
- Countries trade goods **across time**: a deficit is an import of present goods paid for with future goods
- International capital markets let a country **smooth consumption** when income fluctuates and **invest** beyond its own saving
- This approach is **dynamic**, allowing for persistent Current Account surpluses or deficits, contrasting sharply with static trade models (e.g., Heckscher-Ohlin)

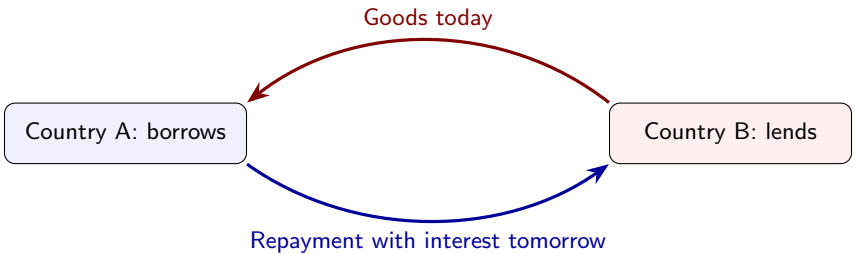
Trade in Time: The Simplest Economic Story

Country A: income low today

A natural disaster lowers output this year, but rebuilding will restore it next year. Without foreign borrowing, today's consumption must fall almost one for one.

Country B: income temporarily high

A temporary export windfall raises national resources now. Instead of spending it all immediately, households save and earn interest.



Takeaway: A deficit exchanges a claim on tomorrow's goods for goods today; a surplus does the reverse. Both countries can gain from smoothing.

Key Concepts

- **Current account:** $CA_t = TB_t + rB_{t-1}$, where $TB_t = Y_t - C_t - I_t - G_t$ is the trade balance and rB_{t-1} net investment income on net foreign assets B_{t-1} held at the end of $t-1$
- **Intertemporal budget constraint (IBC)** – no Ponzi schemes:

$$(1+r)\,B_{t-1} + \sum_{s=t}^{\infty} \frac{TB_s}{(1+r)^{s-t}} = 0$$

  A debtor ($B_{t-1} < 0$) must run future *trade* surpluses whose present value repays its net foreign debt with interest. (Derivation: appendix A.1)
- **Consumption smoothing:** households prefer stable consumption, so they borrow when income is temporarily low and save when it is temporarily high
- **Temporary vs. permanent shocks:** temporary shocks move the current account; permanent shocks move consumption
- **Autarky:** $C_t = Y_t$ every period. **Open economy:** borrow and lend at the world rate r

Two-Period Small Open Economy: Assumptions

- One consumption good, two dates $(1, 2)$.
- Endowments Y_1, Y_2 are known and exogenous.
- The economy can borrow or lend at the given world rate r.
- $B_0 = 0$ initially and $B_2 = 0$ at the end.
- No government or capital accumulation yet.

Households prefer smoother consumption

$$U = u(C_1) + \beta u(C_2)$$

$u' > 0$ and $u'' < 0$: the next unit of consumption is worth more when consumption is low.

Takeaway: An international asset market separates the *timing* of consumption from the timing of income.

Two-Period Model: Sequential Budget Constraints

Period 1: income is either consumed now or saved abroad:

$$C_1 + B_1 = Y_1.$$

Period 2: consume next year's income plus principal and interest:

$$C_2 = Y_2 + (1 + r)B_1.$$

If $B_1 > 0$

The country saves today, so $C_1 < Y_1$; it can afford more consumption next period.

If $B_1 < 0$

The country borrows today, so $C_1 > Y_1$; it repays by reducing consumption tomorrow.

Takeaway: The financial asset B_1 is the bridge between the two budget constraints.

Two Periods Become One Lifetime Budget

Step 1 – solve period 1 for foreign assets:

$$B_1 = Y_1 - C_1.$$

Step 2 – replace B_1 in the second-period constraint:

$$C_2 = Y_2 + (1 + r)(Y_1 - C_1).$$

Step 3 – divide by $1 + r$ and collect consumption:

$$\boxed{C_1 + \frac{C_2}{1 + r} = Y_1 + \frac{Y_2}{1 + r} \equiv W.}$$

Economic interpretation

Lifetime spending cannot exceed lifetime resources. The country can *move* purchasing power through time, but not create unlimited real resources by borrowing.

Takeaway: The interest rate is the relative price of consumption today versus tomorrow.

Optimal Consumption: The Household's Problem

We know *what* the household can afford. Now ask how it chooses the timing of consumption.

$$\max_{C_1, C_2} \quad u(C_1) + \beta u(C_2) \quad \text{s.t.} \quad C_1 + \frac{C_2}{1+r} = W.$$

Lagrangian:

$$\mathcal{L} = u(C_1) + \beta u(C_2) + \lambda \left(W - C_1 - \frac{C_2}{1+r} \right).$$

First-order conditions:

$$u'(C_1) = \lambda, \qquad \beta u'(C_2) = \frac{\lambda}{1+r}.$$

Takeaway: The multiplier λ is the marginal value of one additional unit of lifetime resources.

The Euler Equation: Why Consumption Is Smoothed

Eliminate the multiplier from the two first-order conditions:

$$\boxed{u'(C_1) = \beta(1+r)u'(C_2).}$$

Consider saving one more unit today

The cost is $u'(C_1)$ in lost current utility. The benefit is $1+r$ units of additional future consumption, discounted by patience β: $\beta(1+r)u'(C_2)$.

- If $\beta(1+r) = 1$, consumption is flat: $C_1 = C_2$.
- If $\beta(1+r) > 1$, consumption optimally grows; if below one, it falls.
- The relationship also depends on curvature of u: stronger aversion to consumption fluctuations means greater smoothing.

Takeaway: The intertemporal budget determines *how much*; the Euler equation determines *when*.

Solving the Model: Log Utility

Suppose $u(C) = \ln C$, so marginal utility is $u'(C) = 1/C$. **Step 1 – Euler equation:**

$$\frac{1}{C_1} = \frac{\beta(1+r)}{C_2} \quad \Rightarrow \quad C_2 = \beta(1+r)C_1.$$

Step 2 – insert into the lifetime budget:

$$C_1 + \frac{\beta(1+r)C_1}{1+r} = W \quad \Rightarrow \quad (1+\beta)C_1 = W.$$

Step 3 – solve:

$$\boxed{C_1 = \frac{Y_1 + Y_2/(1+r)}{1+\beta}, \qquad C_2 = \beta(1+r)C_1.}$$

Takeaway: Today's consumption is a share of *lifetime wealth*, not necessarily a share of today's income.

From Optimal Consumption to the Current Account

Current account in period 1 is what the country earns today minus what it consumes today:

$$CA_1 = B_1 = Y_1 - C_1.$$

Substitute the solution:

$$\boxed{CA_1 = \frac{\beta Y_1 - Y_2/(1+r)}{1+\beta}.}$$

Temporary current income windfall

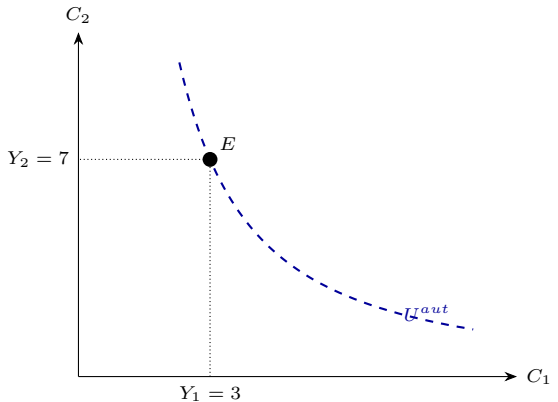
Y_1 rises but Y_2 does not. The country saves part of today's extra income, running a surplus.

Anticipated future income windfall

Y_2 rises. The country can borrow today against expected future resources, running a deficit.

Takeaway: The current account reflects the *gap* between current income and consumption justified by permanent resources.

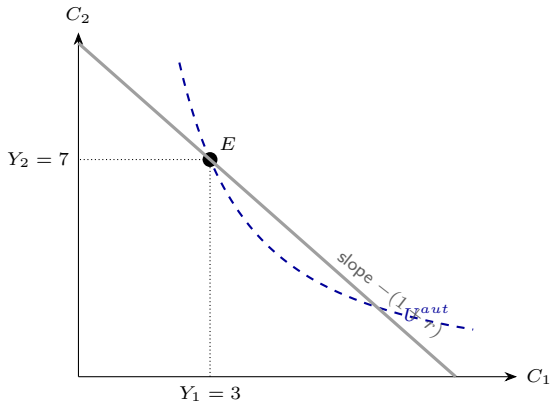
Graphical Intuition 1: The Autarky Endowment



- In autarky there is no international borrowing or lending: consumption is forced to equal the endowment $E = (Y_1, Y_2)$.
- Preferences determine an indifference curve through E.
- Its slope gives the domestic willingness to exchange consumption across time, or the *autarky* rate r^{aut}.

Takeaway: The economy may be forced to consume a very uneven income path, even if it prefers a smoother one.

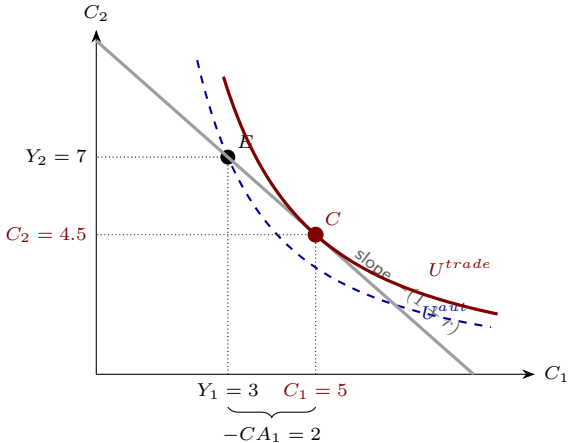
Graphical Intuition 2: Opening to World Capital Markets



- Trade introduces a whole **budget line** through the endowment point E.
- Its slope is $-(1+r)$: sacrificing one unit today yields $1+r$ next period.
- If the world interest rate differs from r^{aut}, the household can reach a higher indifference curve by borrowing or lending.

Takeaway: Opening the capital account expands the set of feasible *consumption profiles*; it does not change the underlying endowment.

Graphical Intuition 3: Optimal Borrowing and the CA



- The optimal bundle C is where the budget line is tangent to the highest attainable indifference curve.
- Illustrated: $E = (3, 7)$, $r = 25\%$, $\beta = .72$; log utility gives $C = (5, 4.5)$.
- The country borrows $C_1 - Y_1 = 2$ today and repays $2(1.25) = 2.5$ tomorrow.
- Thus $CA_1 = 3 - 5 = \mathbf{-2}$.

Takeaway: A current account deficit is the amount of current consumption financed with a claim on future resources.

Responses to Income Shocks: The Current Account as a Shock Absorber

Example: $\beta = 1/(1+r)$, $r = 5\%$, baseline $Y_1 = Y_2 = 100$, so $C_1 = C_2 = 100$ and $CA_1 = 0$.

Shock	(Y_1, Y_2)	$C_1 = C_2$	CA_1	**Intuition**
Temporary fall today	(80, 100)	89.8	-9.8	Borrow to spread the loss over both periods
Permanent fall	(80, 80)	80.0	0.0	Nothing to smooth: consumption adjusts fully
Anticipated rise tomorrow	(100, 120)	109.8	-9.8	Borrow against future income
Temporary rise today	(120, 100)	110.2	$+9.8$	Save the windfall

$C = \frac{(1+r)Y_1 + Y_2}{2+r}$; e.g. $\frac{1.05 \times 80 + 100}{2.05} = 89.8$.

Takeaway: Temporary shocks show up in the current account, permanent shocks in consumption. A country hit by a temporary shock should borrow, not cut consumption one for one.

Trade Balance vs. Current Account

- **Trade balance:** goods and services only, $TB_t = Y_t - C_t - I_t - G_t$
- **Current account:** the trade balance plus net investment income, $CA_t = TB_t + rB_{t-1}$
- **In a steady state** net foreign assets are constant, $CA = 0$, so

$$\overline{TB} = -r\,\overline{B}$$

- A **creditor** ($\overline{B} > 0$) can run a trade deficit forever, financed by its investment income
- A **debtor** ($\overline{B} < 0$) must run a trade surplus forever just to service its debt

Takeaway: The Czech Republic illustrates the second case through equity, not debt: its trade surplus (about 5% of GDP) pays the profits of foreign-owned firms (lecture 2).

Beyond Consumption: Investment Can Also Create Deficits

Imagine a country discovers a profitable oil field or gains access to an exceptionally productive technology.

Without international capital

Firms can invest only if domestic resources are freed by lower consumption. The domestic interest rate may rise to balance saving and investment.

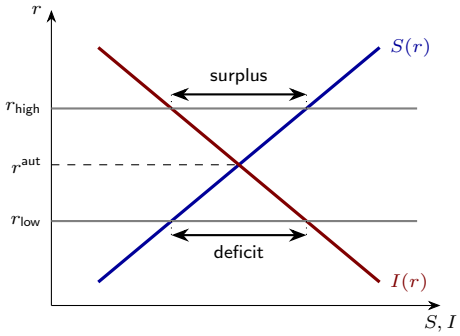
With international capital

Firms can invest more immediately at the world interest rate. If investment rises faster than saving, the current account moves into deficit.

$$\boxed{CA = S - I}$$

Takeaway: A deficit can mean productive investment rather than excessive consumption; its sustainability depends on future returns.

Adding Investment: $CA = S(r) - I(r)$



- Firms invest until the marginal product of capital equals the world interest rate: I falls with r
- The upward-sloping $S(r)$ shown is an illustrative case; saving need not always rise with r
- In autarky $S = I$ at r^{aut}
- If $r > r^{\text{aut}}$: $S > I$, a **surplus**; if $r < r^{\text{aut}}$: a **deficit**
- A productivity boom shifts $I(r)$ right: a deficit that finances investment, repaid from future output

Takeaway: A current account deficit can reflect good news – high returns on domestic investment – not only low saving.

Source: After G. Hall, Brandeis University, Econ 303 notes.

The Infinite Horizon and the "Fundamental" Current Account Equation

- Utility $\sum_{s\geq t}\beta^{s-t}u(C_s)$; with uncertainty the Euler equation is $u'(C_t)=\beta(1+r)\,E_t\,u'(C_{t+1})$
- The IBC (no-Ponzi condition) still binds: $(1+r)B_{t-1}+\sum_{s\geq t}TB_s/(1+r)^{s-t}=0$
- Under certainty and $\beta(1+r)=1$, consumption is flat at permanent income, and the current account is

$$CA_t=(Y_t-\tilde{Y}_t)-(I_t-\tilde{I}_t)-(G_t-\tilde{G}_t),\qquad \tilde{X}_t\equiv\frac{r}{1+r}\sum_{s\geq t}\frac{X_s}{(1+r)^{s-t}}$$

where $\tilde{X}$ is the *permanent* level of X (appendix A.2)

Reading the equation

The country runs a surplus when output is temporarily **above** its permanent level, or investment or government spending temporarily **below** theirs.

Policy reading

A temporary rise in government spending (a war, a natural disaster) should be financed by borrowing abroad – a deficit; a permanent rise needs higher taxes.

Two Large Countries: Who Becomes the Lender?

Country with a low autarky interest rate

Households are relatively patient or current income is high relative to future income. At a higher world rate, they prefer to **save and lend**.

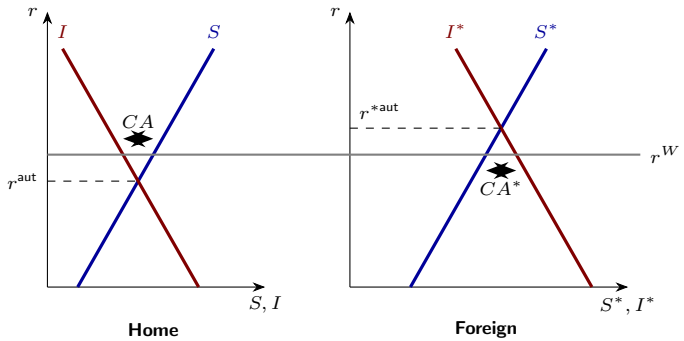
Country with a high autarky interest rate

Households are relatively impatient or expect faster income growth. At a lower world rate, they prefer to **borrow**.

$$r^{aut}_{\text{lender}} < r^{W} < r^{aut}_{\text{borrower}}.$$

Takeaway: The world interest rate coordinates two countries' desired intertemporal trades, just as a goods price coordinates trade across countries.

Two Large Countries: The World Interest Rate



- With large countries, r is **endogenous**
- World market clearing: $CA + CA^* = 0$, i.e. world saving = world investment
- r^W lies between the autarky rates: the patient (low-r^{aut}) country lends
- A shock in one country – a fiscal expansion, a saving glut – moves r^W and spreads abroad

Takeaway: Bernanke's (2005) "global saving glut": higher saving in Asia and oil exporters lowered world interest rates and financed US deficits.

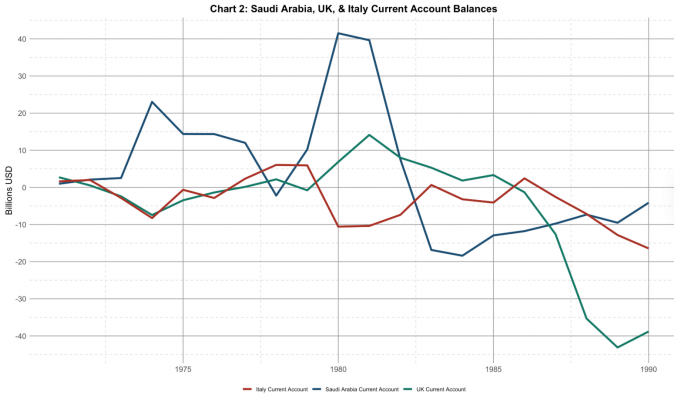
Source: The "Metzler diagram", after Obstfeld and Rogoff (1996), ch. 1; I. Wolden Bache and G. Hall, lecture notes.

Applications: Smoothing, Investing and Puzzles

- **1970s oil shocks:** exporters (Saudi Arabia) ran large surpluses and lent them abroad; importers borrowed to smooth the temporary loss of income (next slide)
- **Norway:** borrowed first – deficits of close to 12% of GDP in the 1970s to finance oil investment – then saved the temporarily high oil income in its fund. The fiscal rule spends only the expected real return (3% since 2017, 4% before): consumption out of *permanent* income
- **Israel, 1990s:** about a million immigrants from the former Soviet Union needed housing and capital; an *investment-driven* deficit that peaked at about 6% of GDP in the mid-1990s
- **US–China:** the US deficit reflects low saving. China's surplus is a **puzzle** for the model: a fast-growing economy should borrow against its future income, not lend – the "allocation puzzle" (Gourinchas and Jeanne 2013)
- **Sovereign debt:** debt is sustainable only if the future trade surpluses the IBC requires are feasible; Greece needed surpluses but ran deficits until 2008

Source: Norges Bank (Ø. Olsen, "Does Norway save too much?", 2018); Jewish Virtual Library, "The Israeli economy 1990–2000"; Gourinchas & Jeanne (2013).

Example: The Oil Shocks of the 1970s



Takeaway: Saudi Arabia saved its windfalls (surpluses of about \$23 bn in 1974 and \$41 bn in 1980) and borrowed once oil prices fell; Italy borrowed after the shocks – smoothing in practice.

Source: Benigno, G. (2025), "Background note on trade and trade imbalances", Substack.

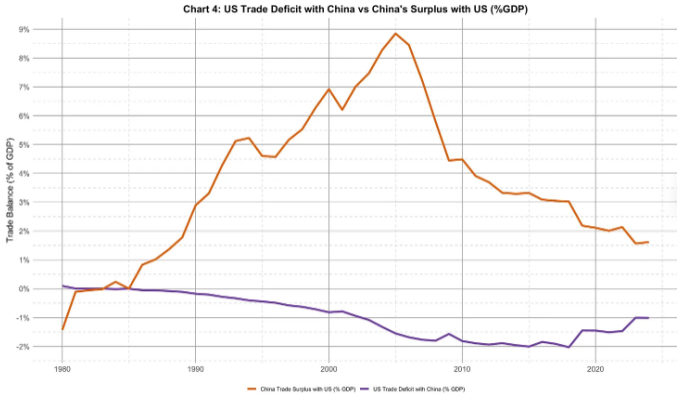
The US and China: Current Account and Trade Balances



Takeaway: China's current account surplus peaked near 10% of GDP in 2007 and was 1–3% for most of the following decade; in 2025 it reached a record $735 billion. The US deficit peaked near 6% of GDP in 2006.

Source: Benigno, G. (2025), "Background note on trade and trade imbalances", Substack; SAFE (February 2026).

The US–China Bilateral Trade Balance



Takeaway: Measured against China's GDP, its surplus with the US fell from about 9% (2006) to under 2% (2024); against US GDP the bilateral deficit is now about 1%. Bilateral balances are not what the model explains – the overall balance is.

Source: Benigno, G. (2025), "Background note on trade and trade imbalances", Substack.

The Persistent US Deficit and Low US Saving

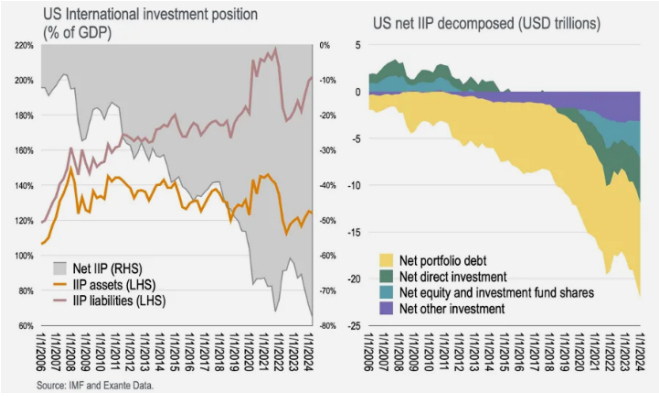
- A deficit almost every year since the early 1980s, widest in 2006 (about 6% of GDP); $1.12 trillion (3.6% of GDP) in 2025 and 3.0% of GDP in Q2 2026
- In the model's terms: national saving below investment, year after year
- US net national saving fell from about 10% of national income in the 1960s to around zero by 2008
- Financed by foreigners' demand for US assets – the dollar's reserve role (lecture 2)



Takeaway: Persistent deficits are hard to read as consumption smoothing; they point to low saving and to the world's demand for safe dollar assets.

Source: BEA, U.S. international transactions (2025 and Q2 2026); FRED (personal saving rate; net national saving, % of GNI).

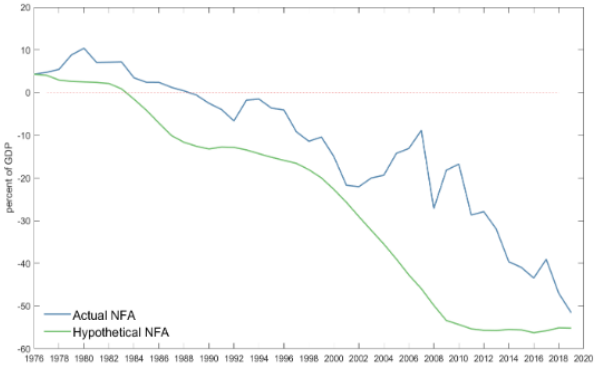
The US International Investment Position



Takeaway: Decades of deficits leave a stock: the US net position fell from about -10% of GDP (2007) to about -80% (2024), mostly because foreign holdings of US debt and equity securities grew.

Source: Exante Data (J. Nordvig), Money Inside Out (Substack), "United States: changing net IIP"; data: IMF.

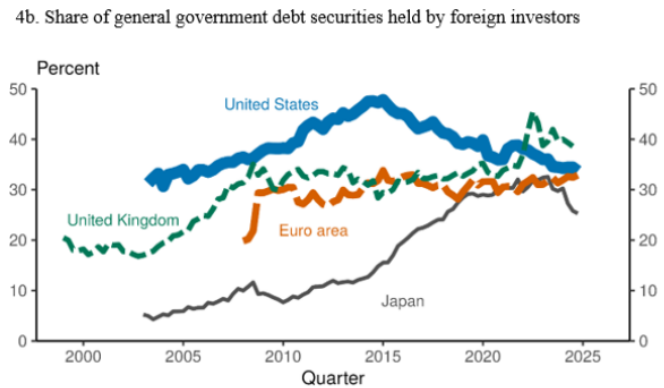
The US NIIP and Valuation Changes



Takeaway: Until about 2010 valuation gains kept the US position far less negative than cumulated deficits imply; by 2019, when this chart ends, the gap had closed. The model's $\Delta B = CA$ ignores this channel (lecture 1).

Source: G. Hall, Brandeis University, Econ 303 notes, "The NIIP paradox"; data: BEA.

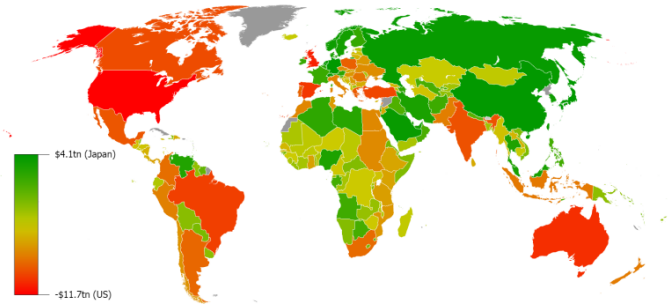
Who Finances the US? Foreign Holdings of Government Debt



Takeaway: Foreigners hold about a third of US government debt, down from almost half in 2014 – similar to the euro area and the UK; the US now relies more on domestic buyers.

Source: Federal Reserve Board (2025), "The international role of the U.S. dollar – 2025 edition", FEDS Notes.

Cumulative Current Account Balances



Takeaway: Over the period shown, the US cumulative deficit reached about $11.7 trillion and Japan's cumulative surplus about $4.1 trillion – the two ends of the colour scale.

Source: Map based on IMF data, reproduced from D. Wales, lecture notes; cumulative period as in the original.

Testing the Model: The Feldstein–Horioka Puzzle

The test (Feldstein and Horioka 1980): regress investment on saving across countries,

$$\left(\frac{I}{Y}\right)_i = \alpha + \beta_{FH}\left(\frac{S}{Y}\right)_i + \varepsilon_i$$

- With perfect capital mobility, domestic saving should flow to the best investments worldwide: β_{FH} near 0
- **Result:** $\beta_{FH} \approx 0.89$ for 16 OECD countries, 1960–74 – saving stayed at home

What happened next

- The coefficient fell over time as capital markets opened
- In the euro area it nearly disappeared: Portugal and Greece ran large deficits as capital flowed in – "the end of the Feldstein–Horioka puzzle?" (Blanchard and Giavazzi 2002)
- A high correlation need not mean low mobility: the IBC alone forces saving and investment to move together in the long run

Source: Feldstein & Horioka (1980), Economic Journal; Blanchard & Giavazzi (2002), Brookings Papers on Economic Activity.

Current Issue: China's Record Surplus – Smoothing or Puzzle?

The facts

- China's current account surplus: a record $735 billion in 2025, about 0.6% of world GDP; the US deficit about 0.9%
- Global imbalances widened again in 2024 and 2025 (IMF)

Ask the model

- Is China's income temporarily high, or are investment opportunities temporarily low?
- Is high household saving a response to an ageing population and weak social insurance?
- What would the model predict if China's growth slowed permanently?

In the two-country log-utility benchmark, the country whose income grows *faster* than the world's should borrow. China grows faster and lends – the allocation puzzle, now at record scale.

Source: SAFE (February 2026); IMF, External Sector Report 2025 and 2026.

Criticisms: A "Real" Model in a "Financial" World

The intertemporal approach explains **why a country wants to borrow or lend**: households smooth consumption, firms invest, and the current account records the net transfer of resources. Its financial critics

(Borio and Disyatat 2011, 2015; Avdjiev, McCauley and Shin 2016) highlight three blind spots:

1. **Net versus gross flows:** the current account hides large, offsetting international financial transactions.
2. **Saving versus financing:** national saving is an *ex post* accounting measure, not a description of the *ex ante* cash needed to spend.
3. **Endogenous money:** commercial banks create deposits when they extend loans; credit need not wait for previously accumulated deposits.

The question for financial stability

Can two countries with the *same* current account balance face very different financial risks? Yes: their banks, funding structures and gross positions may differ greatly.

Takeaway: The current account captures net resource transfers. Financial vulnerability requires additional information about balance sheets and credit.

Criticism 1: Net vs. Gross Capital Flows

The problem: the current account nets transactions that may create very different financial exposures.

Net flows – what the model records

$$CA = \Delta B = -2.$$

A deficit of 2 means foreign liabilities rise *relative to* foreign assets by 2 (ignoring valuation changes). This tells us the size of the net resource transfer, but little about currency, maturity or counterparties.

Gross flows – what financing involves

Suppose residents acquire **100** in foreign assets and non-residents acquire **102** in domestic assets:

$$\Delta A = 100, \qquad \Delta L = 102.$$

$$\Delta B = \Delta A - \Delta L = -2.$$

Gross asset transactions total **202**, despite the tiny net flow.

Takeaway: A nearly balanced current account can coexist with large cross-border positions, leverage and potentially fragile funding.

Criticism 1: Why Gross Flows Matter – Before 2008

Global saving glut (Bernanke 2005)

- Asian and oil-exporting economies generated large **current account surpluses**.
- Their demand for foreign assets may have depressed long-term yields, helping finance US **net borrowing**.
- This is a story about saving, world interest rates and the global allocation of resources.

Global banking glut (Shin 2012)

- European banks raised dollar funding and built large holdings of US mortgage-related securities.
- These **gross dollar positions** were possible despite the euro area's roughly balanced current account.
- When short-term funding became stressed, banks needed dollar liquidity.

Why the distinction matters: a current account balance cannot reveal whether an intermediary has borrowed short in dollars to hold risky long-term assets.

Takeaway: Saving patterns help explain net imbalances; gross-flow and balance-sheet analysis helps explain financial transmission and the 2008 funding crisis.

Source: Bernanke (2005); Shin (2012); Borio & Disyatat (2011); Avdjiev, McCauley & Shin (2016).

Criticism 2: Saving vs. Financing – The Concept

The two concepts are related, but they answer **different questions**.

Saving – an ex post outcome

$$S = Y - C - G.$$

Income not consumed or spent by government, measured in the **national accounts**. The identity

$CA = S - I$ tells us how national saving and investment add up *after* economic transactions have occurred.

Financing – an ex ante mechanism

How does an agent obtain **purchasing power now** to pay for machinery, imports or other expenditures? Possible channels: bank credit, bond issuance, use of deposits, asset sales and cross-border borrowing.

Illustration: a firm may arrange a bank loan *before* its investment creates incomes and the saving recorded in the national accounts.

Takeaway: $CA = S - I$ is an accounting identity, not a description of the financial transactions that make investment possible.

Criticism 2: Why Conflating Them Matters

1. **Misinterpreting causality.** A deficit coincides with investment exceeding national saving; this does not identify which banks, markets or currencies provided the initial liquidity.
2. **Missing the credit cycle.** Rapid credit expansion can occur with a **surplus** (Japan in the late 1980s) or a **deficit** (US before 2008).
3. **Policy risk.** Monitoring saving–investment gaps alone may miss excessive bank leverage, wholesale funding dependence and currency mismatches.

Important qualification

Saving is not the same as financing, but real resources are still scarce. Financial credit can mobilise purchasing power; it cannot remove the intertemporal budget constraint or the external settlement requirement.

Takeaway: The current account describes the net outcome; the financing mechanism explains which constraints bind *during* the adjustment.

Source: Borio & Disyatat (2011, 2015); McLeay, Radia & Thomas (2014).

Criticism 3: Endogenous Money and the Order of Events

Bank lending does not mechanically require a prior transfer of deposits from savers. When a bank makes a loan, it usually creates a matching deposit.

Bank balance sheet, at loan origination	**Assets**	**Liabilities**
New loan to a firm	$+100$	
New deposit of the firm		$+100$

1. The bank extends credit and credits the borrower's account.
2. The borrower spends the deposit; recipients gain income, and payments may require interbank settlement.
3. The national accounts record the resulting investment, income and saving *ex post*.

$$\text{Loan} \Rightarrow \text{Deposit} \Rightarrow \text{Spending} \Rightarrow \text{ex post saving.}$$

Takeaway: Credit creation can precede the saving it helps generate. This is a monetary mechanism omitted from the simple real model.

Source: McLeay, Radia & Thomas (2014), Bank of England, "Money creation in the modern economy".

Criticism 3: Policy Implications of Endogenous Credit

- **Financial elasticity:** banks can expand credit and balance sheets rapidly, sometimes well beyond what current account data suggest.
- **Financial fragility:** credit booms can build maturity mismatches, leverage and foreign-currency exposures *without* a large external deficit.
- **International payments still matter:** domestic loan creation does not itself deliver foreign currency to an exporter abroad. Payments require settlement and corresponding changes in external claims or liabilities.
- **Policy response:** examine gross flows, bank funding currencies, liquidity, maturity mismatch and leverage alongside $CA = S - I$.

Example: a dollar funding squeeze

A European bank holding US securities may face a dollar liquidity shortage even when its home country's current account is close to balance. Its *funding structure*, rather than the aggregate CA, explains the immediate vulnerability.

Takeaway: Financial crises depend on how credit and funding are organised, not simply on whether a country is a net borrower.

Overall Implications: Keep the Model, Add Finance

The original critique recommends three changes in the way we analyse global imbalances and financial crises:

1. **From net to gross flows:** study both the current account and the composition of international assets, liabilities and transactions.
2. **Distinguish saving from financing:** keep $CA = S - I$ as an identity, but ask how households, firms and governments obtain liquidity and credit.
3. **Model banking and endogenous money:** incorporate balance-sheet constraints, currency and maturity mismatch, credit creation and the possibility of funding runs.

Two approaches, two purposes

The intertemporal approach explains *why* economies want to transfer real resources across time. Financial models explain *how* those transfers and much larger gross positions are financed – and where fragility builds up.

Takeaway: The real and financial approaches are complements: neither the resource constraint nor the banking mechanism can be ignored.

What the Real and Financial Approaches Each Explain

Intertemporal approach

Explains incentives to borrow and lend, optimal consumption smoothing, investment opportunities and the net external asset position.

Gross-flow and banking approach

Explains financing chains, asset and liability compositions, currency and maturity mismatch, liquidity shortages and crisis transmission.

Teaching message: Neither view makes the other's accounting identities disappear. The two are *complements*; combining them is essential for understanding contemporary global imbalances and financial crises.

Takeaway: Use the current account to analyse net resource transfers, and gross financial data to analyse financial vulnerability.

Conclusions

- **The core insight:** the current account is the outcome of forward-looking decisions to smooth consumption and to invest. Deficits are not bad in themselves if they smooth a temporary shock or finance productive investment
- **The constraint:** the IBC binds – debtors must run future trade surpluses; creditors can run trade deficits financed by investment income
- **The world interest rate** balances world saving and investment; shocks in large countries spread through it
- **Successes:** oil shocks, Norway's oil fund, investment-driven deficits
- **Puzzles:** saving that stays at home (Feldstein–Horioka); capital flowing from fast-growing China to the slower-growing US (the allocation puzzle)
- **Limits:** a real model in a financial world. For crises, look at gross flows, financing and credit – not only at the current account

Sources and Further Reading

- Avdjiev, S., McCauley, R., & Shin, H. S. (2016). Breaking free of the triple coincidence in international finance. *Economic Policy*, 31(87), 409–451 (BIS WP 524, 2015).
- Benigno, G. (2025). Background note on trade and trade imbalances. Substack.
- Bernanke, B. (2005). The global saving glut and the U.S. current account deficit. Sandridge Lecture, 10 March.
- Blanchard, O., & Giavazzi, F. (2002). Current account deficits in the euro area: The end of the Feldstein–Horioka puzzle? *Brookings Papers on Economic Activity*, 2002(2), 147–209.
- Borio, C., & Disyatat, P. (2011). Global imbalances and the financial crisis: Link or no link? BIS WP 346.
- Borio, C., & Disyatat, P. (2015). Capital flows and the current account: Taking financing (more) seriously. BIS WP 525.
- Feldstein, M., & Horioka, C. (1980). Domestic saving and international capital flows. *Economic Journal*, 90(358), 314–329.
- Felipe, J. (2024). The Feldstein-Horioka Puzzle Revisited: A Behavioral Explanation.
- Gourinchas, P.-O., & Jeanne, O. (2013). Capital flows to developing countries: The allocation puzzle. *Review of Economic Studies*, 80(4), 1484–1515.
- McLeay, M., Radia, A., & Thomas, R. (2014). Money creation in the modern economy. Bank of England *Quarterly Bulletin*, Q1.
- Obstfeld, M., & Rogoff, K. (1995). The intertemporal approach to the current account. In G. Grossman & K. Rogoff (eds.), *Handbook of International Economics*, vol. 3. North-Holland.
- Obstfeld, M., & Rogoff, K. (1996). *Foundations of International Macroeconomics*. MIT Press.
- Olsen, Ø. (2018). Does Norway save too much? Speech, Norges Bank, 27 March.
- Sachs, J. (1981). The current account and macroeconomic adjustment in the 1970s. *Brookings Papers on Economic Activity*, 1981(1), 201–268.
- SAFE (2026). China's balance of payments, 2025 (preliminary). IMF, *External Sector Report* 2025 and 2026. BEA, U.S. international transactions and investment position.
- Lecture notes: E. Kaplan (University of Maryland), `https://econweb.umd.edu/~kaplan/courses/intlmacroslides1.pdf`; D. Wales, `https://danwales.com/teaching/`; G. Hall (Brandeis), Econ 303; I. Wolden Bache; D. Bergholt; D. Donaldson, `https://dave-donaldson.com/teaching/`.

A.1 Deriving the Intertemporal Budget Constraint

Economic question. What is the *lifetime* limit on deficits, rather than the limit in any single year?

Step 1 – rearrange the one-period budget:

$$B_t = (1+r)B_{t-1} + TB_t \quad \Longrightarrow \quad (1+r)B_{t-1} = B_t - TB_t.$$

Step 2 – eliminate next year's assets. Since $B_t = (B_{t+1} - TB_{t+1})/(1+r)$,

$$(1+r)B_{t-1} = -TB_t - \frac{TB_{t+1}}{1+r} + \frac{B_{t+1}}{1+r}.$$

Step 3 – repeat until a terminal date T:

$$(1+r)B_{t-1} = -\underbrace{\left[TB_t + \frac{TB_{t+1}}{1+r} + \cdots + \frac{TB_T}{(1+r)^{T-t}}\right]}_{\text{PV of trade balances}} + \underbrace{\frac{B_T}{(1+r)^{T-t}}}_{\text{discounted terminal assets}}.$$

Step 4 – rule out debt rollovers without repayment and unused terminal wealth. If the discounted terminal asset position tends to zero, then

$$\boxed{(1+r)B_{t-1} + \sum_{s=t}^{\infty} \frac{TB_s}{(1+r)^{s-t}} = 0.}$$

Takeaway: Borrowing today is feasible only if the present value of later trade balances compensates for the initial foreign-asset position.

A.2(a) What Is Permanent Income?

Economic question. If income fluctuates indefinitely, what constant amount could the country consume every period? **Step 1 – write the lifetime resource constraint:**

$$\sum_{s=t}^{\infty} \frac{C_s}{(1+r)^{s-t}} = (1+r)B_{t-1} + \sum_{s=t}^{\infty} \frac{Y_s - I_s - G_s}{(1+r)^{s-t}}.$$

Step 2 – assume certainty, $r > 0$, and $\beta(1+r) = 1$. The Euler equation gives $C_s = C_t$. Then

$$C_t \underbrace{\left(1 + \frac{1}{1+r} + \frac{1}{(1+r)^2} + \cdots\right)}_{(1+r)/r} = (1+r)B_{t-1} + \sum_{s=t}^{\infty} \frac{Y_s - I_s - G_s}{(1+r)^{s-t}}.$$

Step 3 – convert a present value into a constant annuity:

$$\widetilde{X}_t \equiv \frac{r}{1+r} \sum_{s=t}^{\infty} \frac{X_s}{(1+r)^{s-t}}, \qquad \boxed{C_t = rB_{t-1} + \widetilde{Y}_t - \widetilde{I}_t - \widetilde{G}_t}.$$

Takeaway: Permanent income is the *annuity value* of the entire future resource stream, not simply today's income.

A.2(b) Deriving the Fundamental Current Account Equation

Economic question. Why is the current account driven by *temporary* deviations rather than the level of income?

Step 1 – start with the flow identity:

$$CA_t = Y_t - C_t - I_t - G_t + rB_{t-1}.$$

Step 2 – replace consumption by permanent resources (A.2a):

$$CA_t = Y_t - \left[rB_{t-1} + \widetilde{Y}_t - \widetilde{I}_t - \widetilde{G}_t\right] - I_t - G_t + rB_{t-1}.$$

Step 3 – cancel investment income and collect terms:

$$\boxed{CA_t = (Y_t - \widetilde{Y}_t) - (I_t - \widetilde{I}_t) - (G_t - \widetilde{G}_t).}$$

Economic meaning

A surplus arises when current net resources exceed their permanent level.

Takeaway: The household smooths consumption, so the current account absorbs temporary fluctuations in net resources $Y - I - G$.

A.2(c) Shock Timing: Derive the Signs

Use $CA_t = (Y_t - \widetilde{Y}_t) - (I_t - \widetilde{I}_t) - (G_t - \widetilde{G}_t)$, with $r > 0$. **1. One-time output windfall today, $\Delta Y_t = \Delta > 0$:**

$$\Delta \widetilde{Y}_t = \frac{r}{1+r}\Delta \quad \Rightarrow \quad \boxed{\Delta CA_t = \frac{\Delta}{1+r} > 0.}$$

2. Permanent output increase, $\Delta Y_{t+h} = \Delta$ for every $h \geq 0$:

$$\Delta \widetilde{Y}_t = \frac{r}{1+r} \sum_{h=0}^{\infty} \frac{\Delta}{(1+r)^h} = \Delta \quad \Rightarrow \quad \boxed{\Delta CA_t = 0.}$$

3. Future output increase, expected only at $t+1$:

$$\Delta \widetilde{Y}_t = \frac{r\Delta}{(1+r)^2} \quad \Rightarrow \quad \boxed{\Delta CA_t = -\frac{r\Delta}{(1+r)^2} < 0.}$$

4. Temporary investment or government-spending increase today:

$$\boxed{\Delta CA_t = -\frac{\Delta}{1+r} < 0.}$$

Takeaway: News about tomorrow can cause borrowing today, even when today's output has not changed.